

Spin 1/2 particles:

$$u_s(p) \bar{u}_s(p) = \frac{1}{2}(1 + s\gamma_*) (-\not{p})$$

$s = \pm 1$ which specifies the Helicity

The v spinors had

$$v_s(p) \bar{v}_s(p) = \frac{1}{2}(1 - s\gamma_*) (-\not{p})$$

In the massless limit: $u_{-s}(p) = v_s(p)$

$\Rightarrow$ If we consider massless particles we can work w/ $u_s(p)$ and simply flip s for v .

(i) Negative helicity state: $s = -1$

$$u_{-}(p) \bar{u}_{-}(p) = \frac{1}{2}(1 - \gamma_*) (-\not{p})$$

(ii) Twistor (Srednicki) conventions:

$$\epsilon_{12} = \epsilon_{\dot{1}\dot{2}} = -1 \Rightarrow \epsilon_{\alpha\beta} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_{\alpha\dot{\alpha}}^{\mu} \equiv (\mathbb{1}, \sigma^i)$$

$$\epsilon^{\alpha\beta} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\bar{\sigma}^{\mu\dot{\alpha}\alpha} \equiv \epsilon^{\alpha\beta} \epsilon^{\dot{\alpha}\dot{\beta}} \sigma_{\beta\dot{\beta}}^{\mu} = (\mathbb{1}, -\sigma^i)$$

$$\boxed{p_{\alpha\dot{\alpha}} \equiv p_{\mu} \sigma_{\alpha\dot{\alpha}}^{\mu}} \Leftrightarrow p^{\alpha\dot{\alpha}} = \epsilon^{\alpha\beta} \epsilon^{\dot{\alpha}\dot{\beta}} p_{\beta\dot{\beta}} = p_{\mu} \bar{\sigma}^{\mu\dot{\alpha}\alpha}$$

Now, in the Weyl rep of $\text{Cliff}(M^4)$

$$\gamma^{\mu} = \begin{pmatrix} 0 & \sigma^{\mu} \\ \bar{\sigma}^{\mu} & 0 \end{pmatrix}, \quad P_R = \frac{1}{2}(1 - \gamma_*) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Thus,

$$u_{-}(p) \bar{u}_{-}(p) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} (-1) \begin{pmatrix} 0 & \sigma^{\mu} p_{\mu} \\ \bar{\sigma}^{\mu} p_{\mu} & 0 \end{pmatrix}$$

$$u_{-}(p) \bar{u}_{-}(p) = \begin{pmatrix} 0 & -p_{\mu} \sigma^{\mu}_{\alpha\dot{\alpha}} \\ 0 & 0 \end{pmatrix}$$

$$\text{or } u_{-}(p) \bar{u}_{-}(p) = |0 - p_{\alpha\dot{\alpha}}|$$

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$$\text{however } u_\alpha(p) = \begin{pmatrix} \chi_\alpha \\ 0 \end{pmatrix}$$

Here χ_α is a "numerical" spinor and it is not anticommuting.

$$\chi_\alpha = \sqrt{2\omega} \begin{pmatrix} -\sin(\frac{\theta}{2}) e^{-i\phi} \\ +\cos(\frac{\theta}{2}) \end{pmatrix}$$



χ_α : "twistor!"

$$\Rightarrow \bar{u}_\alpha(p) = (0, \chi^\dagger_\alpha), \quad (\chi_\alpha)^\dagger = \bar{\chi}_\alpha$$

Note, all that we are assuming massless spinors so the massless condition

$$p_\mu p^\mu = 0$$

is the basis of our analysis.

$$u(p) \bar{u}(p) = \begin{pmatrix} 0 & \chi_\alpha \bar{\chi}_\alpha \end{pmatrix}$$

$$\therefore p_{\alpha\dot{\alpha}}^{\text{hull}} = -\chi_\alpha \bar{\chi}_{\dot{\alpha}}$$

$$p_{\mu\sigma^{\mu\nu}}_{\alpha\dot{\alpha}} = p_{\alpha\dot{\alpha}} = \begin{pmatrix} -p^0 + p^3 & p^1 - ip^2 \\ p^1 + ip^2 & -p^0 - p^3 \end{pmatrix}$$

$$\det(p_{\alpha\dot{\alpha}}) = -(p^0)^2 + p^2 = 0 = p_\mu p^\mu$$

and so $p_{\alpha\dot{\alpha}}$ has zero eigenvalues, so it can be written as a projection onto the eigenvector corresponding to the non-zero eigenvalue.

This is precisely what

$$p_{\alpha\dot{\alpha}} = -\chi_\alpha \bar{\chi}_{\dot{\alpha}}$$

represents since

$$p_{\alpha\dot{\alpha}} \chi^\alpha = 0$$

Now, let p and k be two bispinors w/ corresponding "twistors" ϕ_α, k_α

$$\langle p k \rangle = \langle \phi k \rangle \equiv \phi^\alpha k_\alpha$$

Note $\bar{u}_+(p) u_- = \langle p k \rangle$
where $u_+(p) = \begin{pmatrix} 0 \\ \phi^\alpha \end{pmatrix} \Rightarrow \bar{u}_+(p) = (\phi^\alpha, 0)$

$$[p k] \equiv \bar{\phi}_{\dot{\alpha}} \bar{k}^{\dot{\alpha}}$$

Thus, $[p k]^* = \langle p k \rangle$. The product of these twistor products:

$$\begin{aligned} [p k] \langle k p \rangle &= \bar{\phi}_{\dot{\alpha}} k^{\dot{\alpha}} k^\alpha \phi_\alpha = \bar{\phi}_{\dot{\alpha}} \phi_\alpha \bar{k}^{\dot{\alpha}} k^\alpha \\ &= p_{\alpha\dot{\alpha}} k^{\dot{\alpha}\alpha} = \sigma^\mu_{\alpha\dot{\alpha}} p_\mu \bar{\sigma}^{\nu\dot{\alpha}\alpha} k_\nu \\ &= p_\mu k_\nu \text{Tr}(\sigma^\mu \bar{\sigma}^\nu) = 2 p^\mu k_\mu \end{aligned}$$

Whats the point? : We can express scattering amplitudes / interactions using

$$\begin{aligned} \bar{u}_+(p) u_- &= \langle p k \rangle &= \phi^\alpha k_\alpha \\ u_-(p) u_+(q) &= [p k] &= \bar{\phi}_{\dot{\alpha}} \bar{k}^{\dot{\alpha}} \end{aligned}$$